AE 153 Agricultural Meteorology & Climatology

cember 2

Time: 2 hours 30mins.

Section A: Answer Question One (1) and Any Other Question.
Each Question Carries 25 Marks

estion

a. Inc le the equipment and units of measurement for the following weather variables [5 Marks]

Soil temperature
i Solar-radiation
ii Evaporation
i Atmospheric pressure
Wind direction

b. W are the importance of the Stevenson screen at a weather station and the relevance of outh and north facing doors? [4 Marks]

c. W is effective rainfall [2 Marks]

d. Li ix (6) factors that influence the determination of effective rainfall [3 Marks]

e. T 1. presents temperature, rainfall pan evaporation and cup anemometer reading data cted over a period one week at DAE weather station. Calculate: [11 Marks]

T 1. Weather data collected at DAE weather station

e	Rainfall (mm)	Max Temp.	Min Temp.	Pan reading (mm)	Anemometer Reading
01/1	5.9	31.0	22.0	10.4	1692.03
02/1	0.0	31.5	21.6	11.2	1763.16
03/1	0.0	30.8	22.2	11.6	1812.91
04/1	0.0	30.5	22.1	4.0	1888.06
05/1	16.5	31.0	23.2	8.4	1929.89
06/1	9.2	31.0	22.2	2.1	1995.03
07/1	32.5	30.8	22.3	8.7	2048.41
					2106.70

i. Number of rainy days
ii. Mean weekly rainfall
iii. Mean daily temperature
iv. Highest temperature range for the week

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Time: 2 hours 30mins.

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Section A: Answer Question One (1) and Any Other Question.
Each Question Carries 25 Marks

estion 1.

a. Indi e the equipment and units of measurement for the following weather variables [5 Marks]

i. Soil temperature
ii. Solar-radiation
iii. Evaporation
iv. Atmospheric pressure
v. Wind direction

b. Wh are the importance of the Stevenson screen at a weather station and the relevance of the uth and north facing doors? [4 Marks]

c. Wh is effective rainfall [2 Marks]

d. Lis ix (6) factors that influence the determination of effective rainfall [3 Marks]

e. Ta 1. presents temperature, rainfall pan evaporation and cup anemometer reading data co ted over a period one week at DAE weather station. Calculate: [11 Marks]

Ta 1. Weather data collected at DAE weather station

Da e	Rainfall (mm)	Max Temp.	Min Temp.	Pan reading (mm)	Anemometer Reading
01/10	5.9	31.0	22.0	10.4	1692.03
02/10	0.0	31.5	21.6	11.2	1763.16
03/10	0.0	30.8	22.2	11.6	1812.91
04/10	0.0	30.5	22.1	4.0	1888.06
05/10	16.5	31.0	23.2	8.4	1929.89
06/10	9.2	31.0	22.2	2.1	1995.03
07/10	32.5	30.8	22.3	8.7	2048.41
					2106.70

i. Number of rainy days
ii. Mean weekly rainfall
iii. Mean daily temperature
iv. Highest temperature range for the week

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v. Mean pan evaporation
vi. Daily wind run
vii. Highest wind speed (m/s)

Question 2

Table 2 presents the monthly rainfall for June 2012, long term mean annual rainfall and corresponding area of influence for each of the rain gauge.

Table 2: Precipitation and corresponding area of gauge influence in a watershed

Rain gauge	Monthly rainfall for June 2012 (mm)	Long term annual rainfall (mm)	Area of influence (A_i)(ha)
P_1	115	1300	250
P_2	156	1550	95
P_3	140	1350	80
P_4	150	1448	220
P_5	130	1288	160
P_6	X	1290	210
P_7	152	1524	150

a. Under what circumstances is important to use the Thiessen polygon method for rainfall analysis [2 Marks]
b. Write an expression for the estimation of missing data using Normal Ratio Method [3 Marks]
c. Estimate the missing rainfall data for June 2012 at location P_6 using the expression in question 2b and the data in Table 2 [6 Marks]
d. Using the simple arithmetic average calculate mean monthly rainfall for June 2012 watershed [3 Marks]
e. Outline the steps you will follow to develop a Thiessen polygon [3 Marks]
f. Write an equation to represent the average precipitation for the watershed that takes into account the area of influence and calculate the average precipitation for the watershed [6 Marks]
g. Compare and comment on your answers from question 3d and 3f [2 Marks]

Question 3

a. Outline five (5) the importance of radiation [5 Marks]
b. Which of the layer of the atmosphere is closest to the earth? What are the characteristic this layer? Troposphere [3 Marks]
c. What are the causes of global climate change? [4 Marks]
d. What are the likely impacts of climate change? [4 Marks]
e. Outline three (3), climate change related risk in the field of agriculture. For each stated give four (4) examples of the possible impact on the environment [9 Marks]

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Drought
Flood
Airport

f. Table 2 presents temperature, rainfall and cup anemometer data collected
* 10 days at DAE weather station. Calculate:
  i. Daily wind run
  ii. Total 10 day rainfall
  iii. Mean daily temperature
  iv. Temperature range for the 10 days

Question 2

a. List the five (5) layers of the atmosphere
b. Which of the layers listed in question 2b is closest to the earth? What are the characteristics of this layer?
c. What are the causes of global climate change?
d. What are the likely impacts of climate change?
e. Outline three (3), climate change related risk in the field of agriculture. For each risk give four (4) examples of the possible impact on the environment

Question 3

Figure 1 illustrates the location of six rain gauges in a watershed. Table 3 presents the monthly rainfall for May 2011, long term mean annual rainfall and the area of influence at each rain gauge locations.

Table 3: Precipitation and corresponding area of gauge influence in a watershed

Rain gauge	Monthly rainfall for May 2011 (mm)	Long term mean annual rainfall (mm)	Area of influence (A_i)(ha)
P_1	x	1368	260
P_2	156	1324	58
P_3	149	1300	50
P_4	158	1348	250
P_5	148	1288	62
P_6	152	1290	290

a. Write an expression for the estimation of missing data using Normal Ratio Method [3 Marks]
b. Estimate the missing rainfall data for May 2011 at location P_1 using the expression in question 2a [6 Marks]
c. Using the simple arithmetic average calculate mean monthly rainfall for May 2011 in the watershed [3 Marks]
d. Sketch the Thiessen polygon marking out the area of influence of the different rain gauges [4 Marks]
e. Write an equation to represent the average precipitation for the watershed that takes account the area of influence and calculate the average precipitation for the watershed [6 Marks]
f. Compare and comment on your answers from question 3c and d [3 Marks]

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b. List four (4) benefits of radiation [2 Marks]
c. Explain the four (4) ways by which solar radiation passing through the atmosphere can be influenced [8 Marks]
d. What is net radiation? Write an expression for the net radiation and explain the terms involved [5 Marks]
e. Copy and complete Table 4 by indicating for each pair of surface (i.e. I to IV), which surface generally will have high or low albedo [8 Marks]

Table 4: Surfaces and their relative albedo level

	Surface	Albedo level	
		High	Low
I.	Water		
	Soil		
II.	Rain forest		
	Grass		
III.	Wet dark soil		
	Dry dark soil		
IV.	Ploughed land		
	Bare land		

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c. Sketch the Thiessen polygon showing the areas of influence of the different rain gauges.

d. Table 1 presents the precipitation and the corresponding area of influence around of the rain gauge locations. Write an equation to represent the average precipitation for the watershed and calculate the average precipitation for the watershed.

Table 1: Precipitation and corresponding area of gauge influence in a watershed

Rain gauge	Precipitation (mm)	Area of influence (ha)
P_1	56	260
P_2	40	58
P_3	44	50
P_4	52	250
P_5	48	62
P_6	52	290

Question 2

a. Define the following:
   i. Weather [1 Marks]
   ii. Agrometeorology [1 Marks]

b. What is the troposphere and its functions? [3 Marks]

c. Briefly explain the following:
   i. Climate change [2 Marks]
   ii. Global warming [2 Marks]
   iii. Greenhouse gas [2 Marks]

d. Outline two (2) each, climate change related risk in the field of agriculture and water resources. For each stated risk give four (4) examples of the possible impact on the environment [8 Marks]

e. Explain briefly the two (2) basic approaches to limiting climate change or its impact. In each case give two (2) examples of how this can be done [6 Marks]

Question 3

a. Indicate the equipment and units of measurement for the following weather variables [4 Marks]
   i. Wind direction
   ii. Sunshine
   iii. Atmospheric pressure
   iv. Solar-radiation

b. List six (6) factors that influence seasonal temperature variations [3 Marks]

c. Outline six (6) relevance of soil temperature in the field of agriculture [3 Marks]

d. Briefly explain the direct and indirect methods of measuring relative humidity [4 Marks]

e. What are the importance of the Stevenson screen at a weather station and the relevance of the south and north facing doors? [3 Marks]

f. Table 1 presents temperature, rainfall and pan evaporation data collected over a period of [illegible] at AE weather station. Calculate: [8 Marks]

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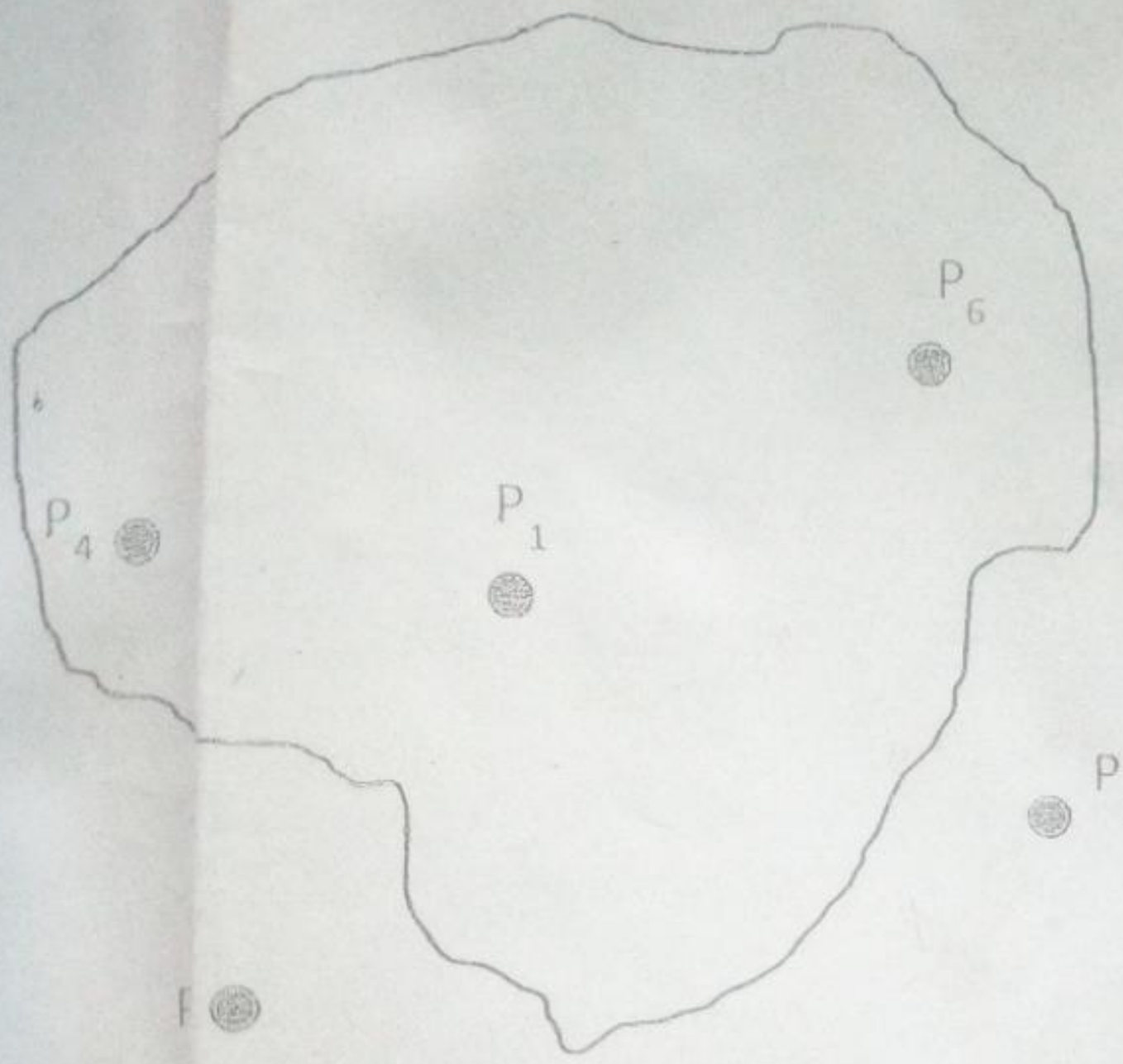


Figure l:ocation of rain gauges in a watershed

stion 4

1. Define tfollowing: [2 Marks]
   i. Wther [2 Marks]
   ii. Gnate [2 Marks]
   iii. Aometeorology
   What are two main factors that determines the day length, seasons and climate variabil of the earth [2 Marks]
   Outline three primary steps in the formation of precipitation [4 Marks]
   What are characteristics of precipitation resulting from warm and cold fronts? [3 Marks]
2. Explain following terminologies used in representing rainfall data [6 Marks]
   i. mulative mass curve
   ii. nfall hyetograph
   iii. hyetal Maps
   What is recording rain gauge? Give two (2) advantages of the recording rain gauge over [4 Marks].
   [2 Marks]